

Cabling Guide — 64× B200 AI Cluster

Rev 0.2 · 2026-07-19

Generated from Markdown — edit the .md source and regenerate (python3 docs/md2pdf.py)

Contents

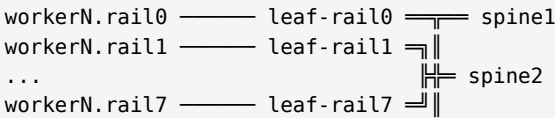
- 1 Design rules (SPEC §5) 3
- 2 Summary counts 4
- 3 Host ↔ rail leaf (GPU fabric) 5
 - 3.1 Install checklist (rail {i}) 6
- 4 Leaf ↔ spine 8
- 5 OOB / mgmt plane (7010TX-48) 10
 - 5.1 BMC (VLAN 20) 10
 - 5.2 Server mgmt0 — OS/PXE (VLAN 10) 10
 - 5.3 Switch Management1 — ZTP (VLAN 20) 11
 - 5.4 OOB MLAG peer-link 12
- 6 Labeling convention 13
- 7 Acceptance 14
- 8 Revision 15

Source of truth: NetBox (seed/site.yaml (offline))
Generated: 2026-07-19 13:35 UTC
Regenerate: python3 bootstrap/netbox/scripts/export_cabling.py

Do not maintain this file by hand. Update NetBox (or bootstrap/netbox/seed/site.yaml pre-go-live), then re-export.

1 Design rules (SPEC §5)

Rule	Practice
Rail identity	<code>workerXX.rail{i}</code> → <code>leaf-rail{i}</code> for all workers
Host media	400G DAC QSFP-DD (short; leaf in same/near rack)
Leaf↔spine	400G AOC/fiber; 8 uplinks/leaf (4 per spine)
OOB	BMC → 7010TX-48 only; no RoCE on OOB
Spines	Prefer BOOT rack
Leaves	Split GPU-1 (rail0–3) / GPU-2 (rail4–7)



2 Summary counts

Class	Cables
Host ↔ rail leaf (8 workers × 8 rails)	64
Rail leaf ↔ spine	64
BMC ↔ OOB (VLAN 20)	14
Server mgmt0 ↔ OOB (VLAN 10)	15
Switch Ma1 ↔ OOB (VLAN 20, ZTP)	10
OOB MLAG peer-link	2
Total	169

3 Host ↔ rail leaf (GPU fabric)

Port map: **worker index 1..8 → leaf Ethernet1..8.**

Label	A device	A port	B device	B port	Type	Status
R0-W01	worker01	rail0	leaf-rail0	Ethernet1	dac-qsfpdd-400g	planned
R0-W02	worker02	rail0	leaf-rail0	Ethernet2	dac-qsfpdd-400g	planned
R0-W03	worker03	rail0	leaf-rail0	Ethernet3	dac-qsfpdd-400g	planned
R0-W04	worker04	rail0	leaf-rail0	Ethernet4	dac-qsfpdd-400g	planned
R0-W05	worker05	rail0	leaf-rail0	Ethernet5	dac-qsfpdd-400g	planned
R0-W06	worker06	rail0	leaf-rail0	Ethernet6	dac-qsfpdd-400g	planned
R0-W07	worker07	rail0	leaf-rail0	Ethernet7	dac-qsfpdd-400g	planned
R0-W08	worker08	rail0	leaf-rail0	Ethernet8	dac-qsfpdd-400g	planned
R1-W01	worker01	rail1	leaf-rail1	Ethernet1	dac-qsfpdd-400g	planned
R1-W02	worker02	rail1	leaf-rail1	Ethernet2	dac-qsfpdd-400g	planned
R1-W03	worker03	rail1	leaf-rail1	Ethernet3	dac-qsfpdd-400g	planned
R1-W04	worker04	rail1	leaf-rail1	Ethernet4	dac-qsfpdd-400g	planned
R1-W05	worker05	rail1	leaf-rail1	Ethernet5	dac-qsfpdd-400g	planned
R1-W06	worker06	rail1	leaf-rail1	Ethernet6	dac-qsfpdd-400g	planned
R1-W07	worker07	rail1	leaf-rail1	Ethernet7	dac-qsfpdd-400g	planned
R1-W08	worker08	rail1	leaf-rail1	Ethernet8	dac-qsfpdd-400g	planned
R2-W01	worker01	rail2	leaf-rail2	Ethernet1	dac-qsfpdd-400g	planned
R2-W02	worker02	rail2	leaf-rail2	Ethernet2	dac-qsfpdd-400g	planned
R2-W03	worker03	rail2	leaf-rail2	Ethernet3	dac-qsfpdd-400g	planned
R2-W04	worker04	rail2	leaf-rail2	Ethernet4	dac-qsfpdd-400g	planned
R2-W05	worker05	rail2	leaf-rail2	Ethernet5	dac-qsfpdd-400g	planned
R2-W06	worker06	rail2	leaf-rail2	Ethernet6	dac-qsfpdd-400g	planned
R2-W07	worker07	rail2	leaf-rail2	Ethernet7	dac-qsfpdd-400g	planned
R2-W08	worker08	rail2	leaf-rail2	Ethernet8	dac-qsfpdd-400g	planned
R3-W01	worker01	rail3	leaf-rail3	Ethernet1	dac-qsfpdd-400g	planned
R3-W02	worker02	rail3	leaf-rail3	Ethernet2	dac-qsfpdd-400g	planned
R3-W03	worker03	rail3	leaf-rail3	Ethernet3	dac-qsfpdd-400g	planned
R3-W04	worker04	rail3	leaf-rail3	Ethernet4	dac-qsfpdd-400g	planned
R3-W05	worker05	rail3	leaf-rail3	Ethernet5	dac-qsfpdd-400g	planned
R3-W06	worker06	rail3	leaf-rail3	Ethernet6	dac-qsfpdd-400g	planned
R3-W07	worker07	rail3	leaf-rail3	Ethernet7	dac-qsfpdd-400g	planned
R3-W08	worker08	rail3	leaf-rail3	Ethernet8	dac-qsfpdd-400g	planned

Label	A device	A port	B device	B port	Type	Status
R4-W01	worker01	rail4	leaf-rail4	Ethernet1	dac-qsfpdd-400g	planned
R4-W02	worker02	rail4	leaf-rail4	Ethernet2	dac-qsfpdd-400g	planned
R4-W03	worker03	rail4	leaf-rail4	Ethernet3	dac-qsfpdd-400g	planned
R4-W04	worker04	rail4	leaf-rail4	Ethernet4	dac-qsfpdd-400g	planned
R4-W05	worker05	rail4	leaf-rail4	Ethernet5	dac-qsfpdd-400g	planned
R4-W06	worker06	rail4	leaf-rail4	Ethernet6	dac-qsfpdd-400g	planned
R4-W07	worker07	rail4	leaf-rail4	Ethernet7	dac-qsfpdd-400g	planned
R4-W08	worker08	rail4	leaf-rail4	Ethernet8	dac-qsfpdd-400g	planned
R5-W01	worker01	rail5	leaf-rail5	Ethernet1	dac-qsfpdd-400g	planned
R5-W02	worker02	rail5	leaf-rail5	Ethernet2	dac-qsfpdd-400g	planned
R5-W03	worker03	rail5	leaf-rail5	Ethernet3	dac-qsfpdd-400g	planned
R5-W04	worker04	rail5	leaf-rail5	Ethernet4	dac-qsfpdd-400g	planned
R5-W05	worker05	rail5	leaf-rail5	Ethernet5	dac-qsfpdd-400g	planned
R5-W06	worker06	rail5	leaf-rail5	Ethernet6	dac-qsfpdd-400g	planned
R5-W07	worker07	rail5	leaf-rail5	Ethernet7	dac-qsfpdd-400g	planned
R5-W08	worker08	rail5	leaf-rail5	Ethernet8	dac-qsfpdd-400g	planned
R6-W01	worker01	rail6	leaf-rail6	Ethernet1	dac-qsfpdd-400g	planned
R6-W02	worker02	rail6	leaf-rail6	Ethernet2	dac-qsfpdd-400g	planned
R6-W03	worker03	rail6	leaf-rail6	Ethernet3	dac-qsfpdd-400g	planned
R6-W04	worker04	rail6	leaf-rail6	Ethernet4	dac-qsfpdd-400g	planned
R6-W05	worker05	rail6	leaf-rail6	Ethernet5	dac-qsfpdd-400g	planned
R6-W06	worker06	rail6	leaf-rail6	Ethernet6	dac-qsfpdd-400g	planned
R6-W07	worker07	rail6	leaf-rail6	Ethernet7	dac-qsfpdd-400g	planned
R6-W08	worker08	rail6	leaf-rail6	Ethernet8	dac-qsfpdd-400g	planned
R7-W01	worker01	rail7	leaf-rail7	Ethernet1	dac-qsfpdd-400g	planned
R7-W02	worker02	rail7	leaf-rail7	Ethernet2	dac-qsfpdd-400g	planned
R7-W03	worker03	rail7	leaf-rail7	Ethernet3	dac-qsfpdd-400g	planned
R7-W04	worker04	rail7	leaf-rail7	Ethernet4	dac-qsfpdd-400g	planned
R7-W05	worker05	rail7	leaf-rail7	Ethernet5	dac-qsfpdd-400g	planned
R7-W06	worker06	rail7	leaf-rail7	Ethernet6	dac-qsfpdd-400g	planned
R7-W07	worker07	rail7	leaf-rail7	Ethernet7	dac-qsfpdd-400g	planned
R7-W08	worker08	rail7	leaf-rail7	Ethernet8	dac-qsfpdd-400g	planned

3.1 Install checklist (rail {i})

For each rail leaf `leaf-rail{i}`:

1. Confirm leaf is powered and Management1 on OOB.
2. Patch Ethernet1...Ethernet8 to worker01...worker08 port rail{i} using labels R{i}-W01 ... R{i}-W08.

3. Dress DACs for service loops; no sharp bend radius < manufacturer min.
 4. Validate link LEDs both ends; record serials in NetBox.
-

4 Leaf ↔ spine

Default: leaf ports **Ethernet17–24** uplinks; spine ports blocked by rail (4 ports/rail/spine).

Label	A device	A port	B device	B port	Type	Status
L0S1-U1	leaf-rail0	Ethernet17	spine1	Ethernet1	aoc-qsfpdd-400g	planned
L0S1-U3	leaf-rail0	Ethernet19	spine1	Ethernet2	aoc-qsfpdd-400g	planned
L0S1-U5	leaf-rail0	Ethernet21	spine1	Ethernet3	aoc-qsfpdd-400g	planned
L0S1-U7	leaf-rail0	Ethernet23	spine1	Ethernet4	aoc-qsfpdd-400g	planned
L0S2-U2	leaf-rail0	Ethernet18	spine2	Ethernet1	aoc-qsfpdd-400g	planned
L0S2-U4	leaf-rail0	Ethernet20	spine2	Ethernet2	aoc-qsfpdd-400g	planned
L0S2-U6	leaf-rail0	Ethernet22	spine2	Ethernet3	aoc-qsfpdd-400g	planned
L0S2-U8	leaf-rail0	Ethernet24	spine2	Ethernet4	aoc-qsfpdd-400g	planned
L1S1-U1	leaf-rail1	Ethernet17	spine1	Ethernet5	aoc-qsfpdd-400g	planned
L1S1-U3	leaf-rail1	Ethernet19	spine1	Ethernet6	aoc-qsfpdd-400g	planned
L1S1-U5	leaf-rail1	Ethernet21	spine1	Ethernet7	aoc-qsfpdd-400g	planned
L1S1-U7	leaf-rail1	Ethernet23	spine1	Ethernet8	aoc-qsfpdd-400g	planned
L1S2-U2	leaf-rail1	Ethernet18	spine2	Ethernet5	aoc-qsfpdd-400g	planned
L1S2-U4	leaf-rail1	Ethernet20	spine2	Ethernet6	aoc-qsfpdd-400g	planned
L1S2-U6	leaf-rail1	Ethernet22	spine2	Ethernet7	aoc-qsfpdd-400g	planned
L1S2-U8	leaf-rail1	Ethernet24	spine2	Ethernet8	aoc-qsfpdd-400g	planned
L2S1-U1	leaf-rail2	Ethernet17	spine1	Ethernet9	aoc-qsfpdd-400g	planned
L2S1-U3	leaf-rail2	Ethernet19	spine1	Ethernet10	aoc-qsfpdd-400g	planned
L2S1-U5	leaf-rail2	Ethernet21	spine1	Ethernet11	aoc-qsfpdd-400g	planned
L2S1-U7	leaf-rail2	Ethernet23	spine1	Ethernet12	aoc-qsfpdd-400g	planned
L2S2-U2	leaf-rail2	Ethernet18	spine2	Ethernet9	aoc-qsfpdd-400g	planned
L2S2-U4	leaf-rail2	Ethernet20	spine2	Ethernet10	aoc-qsfpdd-400g	planned
L2S2-U6	leaf-rail2	Ethernet22	spine2	Ethernet11	aoc-qsfpdd-400g	planned
L2S2-U8	leaf-rail2	Ethernet24	spine2	Ethernet12	aoc-qsfpdd-400g	planned
L3S1-U1	leaf-rail3	Ethernet17	spine1	Ethernet13	aoc-qsfpdd-400g	planned
L3S1-U3	leaf-rail3	Ethernet19	spine1	Ethernet14	aoc-qsfpdd-400g	planned
L3S1-U5	leaf-rail3	Ethernet21	spine1	Ethernet15	aoc-qsfpdd-400g	planned
L3S1-U7	leaf-rail3	Ethernet23	spine1	Ethernet16	aoc-qsfpdd-400g	planned
L3S2-U2	leaf-rail3	Ethernet18	spine2	Ethernet13	aoc-qsfpdd-400g	planned
L3S2-U4	leaf-rail3	Ethernet20	spine2	Ethernet14	aoc-qsfpdd-400g	planned
L3S2-U6	leaf-rail3	Ethernet22	spine2	Ethernet15	aoc-qsfpdd-400g	planned
L3S2-U8	leaf-rail3	Ethernet24	spine2	Ethernet16	aoc-qsfpdd-400g	planned

Label	A device	A port	B device	B port	Type	Status
L4S1-U1	leaf-rail4	Ethernet17	spine1	Ethernet17	aoc-qsfpdd-400g	planned
L4S1-U3	leaf-rail4	Ethernet19	spine1	Ethernet18	aoc-qsfpdd-400g	planned
L4S1-U5	leaf-rail4	Ethernet21	spine1	Ethernet19	aoc-qsfpdd-400g	planned
L4S1-U7	leaf-rail4	Ethernet23	spine1	Ethernet20	aoc-qsfpdd-400g	planned
L4S2-U2	leaf-rail4	Ethernet18	spine2	Ethernet17	aoc-qsfpdd-400g	planned
L4S2-U4	leaf-rail4	Ethernet20	spine2	Ethernet18	aoc-qsfpdd-400g	planned
L4S2-U6	leaf-rail4	Ethernet22	spine2	Ethernet19	aoc-qsfpdd-400g	planned
L4S2-U8	leaf-rail4	Ethernet24	spine2	Ethernet20	aoc-qsfpdd-400g	planned
L5S1-U1	leaf-rail5	Ethernet17	spine1	Ethernet21	aoc-qsfpdd-400g	planned
L5S1-U3	leaf-rail5	Ethernet19	spine1	Ethernet22	aoc-qsfpdd-400g	planned
L5S1-U5	leaf-rail5	Ethernet21	spine1	Ethernet23	aoc-qsfpdd-400g	planned
L5S1-U7	leaf-rail5	Ethernet23	spine1	Ethernet24	aoc-qsfpdd-400g	planned
L5S2-U2	leaf-rail5	Ethernet18	spine2	Ethernet21	aoc-qsfpdd-400g	planned
L5S2-U4	leaf-rail5	Ethernet20	spine2	Ethernet22	aoc-qsfpdd-400g	planned
L5S2-U6	leaf-rail5	Ethernet22	spine2	Ethernet23	aoc-qsfpdd-400g	planned
L5S2-U8	leaf-rail5	Ethernet24	spine2	Ethernet24	aoc-qsfpdd-400g	planned
L6S1-U1	leaf-rail6	Ethernet17	spine1	Ethernet25	aoc-qsfpdd-400g	planned
L6S1-U3	leaf-rail6	Ethernet19	spine1	Ethernet26	aoc-qsfpdd-400g	planned
L6S1-U5	leaf-rail6	Ethernet21	spine1	Ethernet27	aoc-qsfpdd-400g	planned
L6S1-U7	leaf-rail6	Ethernet23	spine1	Ethernet28	aoc-qsfpdd-400g	planned
L6S2-U2	leaf-rail6	Ethernet18	spine2	Ethernet25	aoc-qsfpdd-400g	planned
L6S2-U4	leaf-rail6	Ethernet20	spine2	Ethernet26	aoc-qsfpdd-400g	planned
L6S2-U6	leaf-rail6	Ethernet22	spine2	Ethernet27	aoc-qsfpdd-400g	planned
L6S2-U8	leaf-rail6	Ethernet24	spine2	Ethernet28	aoc-qsfpdd-400g	planned
L7S1-U1	leaf-rail7	Ethernet17	spine1	Ethernet29	aoc-qsfpdd-400g	planned
L7S1-U3	leaf-rail7	Ethernet19	spine1	Ethernet30	aoc-qsfpdd-400g	planned
L7S1-U5	leaf-rail7	Ethernet21	spine1	Ethernet31	aoc-qsfpdd-400g	planned
L7S1-U7	leaf-rail7	Ethernet23	spine1	Ethernet32	aoc-qsfpdd-400g	planned
L7S2-U2	leaf-rail7	Ethernet18	spine2	Ethernet29	aoc-qsfpdd-400g	planned
L7S2-U4	leaf-rail7	Ethernet20	spine2	Ethernet30	aoc-qsfpdd-400g	planned
L7S2-U6	leaf-rail7	Ethernet22	spine2	Ethernet31	aoc-qsfpdd-400g	planned
L7S2-U8	leaf-rail7	Ethernet24	spine2	Ethernet32	aoc-qsfpdd-400g	planned

5 OOB / mgmt plane (7010TX-48)

Rack	OOB switch
GPU-1, BOOT	oob-sw1
GPU-2, STOR	oob-sw2

Port allocation per OOB switch, in order: **BMC (VLAN 20)** → **server mgmt0 (VLAN 10)** → **switch Ma1 (VLAN 20)**; SFP28 49–50 = MLAG peer-link. Full design: docs/network.md §4.

5.1 BMC (VLAN 20)

Label	A device	A port	B device	B port	Type	Status
00B-cp01	cp01	bmc	oob-sw1	Ethernet1	cat6	planned
00B-cp02	cp02	bmc	oob-sw1	Ethernet2	cat6	planned
00B-cp03	cp03	bmc	oob-sw1	Ethernet3	cat6	planned
00B-util01	util01	bmc	oob-sw1	Ethernet4	cat6	planned
00B-util02	util02	bmc	oob-sw1	Ethernet5	cat6	planned
00B-util03	util03	bmc	oob-sw1	Ethernet6	cat6	planned
00B-worker01	worker01	bmc	oob-sw1	Ethernet7	cat6	planned
00B-worker02	worker02	bmc	oob-sw1	Ethernet8	cat6	planned
00B-worker03	worker03	bmc	oob-sw1	Ethernet9	cat6	planned
00B-worker04	worker04	bmc	oob-sw1	Ethernet10	cat6	planned
00B-worker05	worker05	bmc	oob-sw2	Ethernet1	cat6	planned
00B-worker06	worker06	bmc	oob-sw2	Ethernet2	cat6	planned
00B-worker07	worker07	bmc	oob-sw2	Ethernet3	cat6	planned
00B-worker08	worker08	bmc	oob-sw2	Ethernet4	cat6	planned

5.2 Server mgmt0 — OS/PXE (VLAN 10)

Label	A device	A port	B device	B port	Type	Status
MGMT-cp01	cp01	mgmt0	oob-sw1	Ethernet12	cat6	planned
MGMT-cp02	cp02	mgmt0	oob-sw1	Ethernet13	cat6	planned
MGMT-cp03	cp03	mgmt0	oob-sw1	Ethernet14	cat6	planned
MGMT-seed01	seed01	mgmt0	oob-sw1	Ethernet11	cat6	planned

Label	A device	A port	B device	B port	Type	Status
MGMT-util01	util01	mgmt0	oob-sw1	Ethernet15	cat6	planned
MGMT-util02	util02	mgmt0	oob-sw1	Ethernet16	cat6	planned
MGMT-util03	util03	mgmt0	oob-sw1	Ethernet17	cat6	planned
MGMT-worker01	worker01	mgmt0	oob-sw1	Ethernet18	cat6	planned
MGMT-worker02	worker02	mgmt0	oob-sw1	Ethernet19	cat6	planned
MGMT-worker03	worker03	mgmt0	oob-sw1	Ethernet20	cat6	planned
MGMT-worker04	worker04	mgmt0	oob-sw1	Ethernet21	cat6	planned
MGMT-worker05	worker05	mgmt0	oob-sw2	Ethernet5	cat6	planned
MGMT-worker06	worker06	mgmt0	oob-sw2	Ethernet6	cat6	planned
MGMT-worker07	worker07	mgmt0	oob-sw2	Ethernet7	cat6	planned
MGMT-worker08	worker08	mgmt0	oob-sw2	Ethernet8	cat6	planned

5.3 Switch Management1 — ZTP (VLAN 20)

Label	A device	A port	B device	B port	Type	Status
MA1-leaf-rail0	leaf-rail0	Management1	oob-sw1	Ethernet24	cat6	planned
MA1-leaf-rail1	leaf-rail1	Management1	oob-sw1	Ethernet25	cat6	planned
MA1-leaf-rail2	leaf-rail2	Management1	oob-sw1	Ethernet26	cat6	planned
MA1-leaf-rail3	leaf-rail3	Management1	oob-sw1	Ethernet27	cat6	planned
MA1-leaf-rail4	leaf-rail4	Management1	oob-sw2	Ethernet9	cat6	planned
MA1-leaf-rail5	leaf-rail5	Management1	oob-sw2	Ethernet10	cat6	planned
MA1-leaf-rail6	leaf-rail6	Management1	oob-sw2	Ethernet11	cat6	planned
MA1-leaf-rail7	leaf-rail7	Management1	oob-sw2	Ethernet12	cat6	planned
MA1-spine1	spine1	Management1	oob-sw1	Ethernet22	cat6	planned
MA1-spine2	spine2	Management1	oob-sw1	Ethernet23	cat6	planned

5.4 OOB MLAG peer-link

Label	A device	A port	B device	B port	Type	Status
00B-PEER-1	oob-sw1	Ethernet49	oob-sw2	Ethernet49	dac-25g-5m	planned
00B-PEER-2	oob-sw1	Ethernet50	oob-sw2	Ethernet50	dac-25g-5m	planned

6 Labeling convention

Pattern	Meaning
R{rail}-W{nn}	Host fabric
L{rail}S{spine}-U{n}	Leaf-spine uplink
00B-{device}	BMC
MGMT-{device}	Server mgmt0 (OS/PXE)
MA1-{device}	Switch Management1 (ZTP)
00B-PEER-{n}	OOB MLAG peer-link

Print both ends; enter QR/barcode into NetBox cable `label` field on install (status → connected).

7 Acceptance

- ☐ NetBox cable status **connected** matches physical light
 - ☐ `export_inventory.py` shows all mgmt MACs after neighbor discovery (optional)
 - ☐ No host rail cable lands on wrong rail leaf (audit by label)
 - ☐ OOB-only path from jump to every BMC
-

8 Revision

Version	Date	Notes
0.1	2026-07-18	Initial export format
0.2	2026-07-19	1. mgmt0 (VLAN 10), switch Ma1 (ZTP), OOB MLAG peer-link